

Weakness as Local Evidence: A Systems Theory for Degeneracy, Refinement DAGs, and Continual Adaptation, with Transformers as a Case Study

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Abstract

We propose a systems-level formalism for continual learning in nonstationary environments that unifies three themes: (1) the local-learning-coefficient (LLC) view of degeneracy from singular learning theory (SLT); (2) “weakness” as an Occam-like bias formalized in thermodynamic quantales; and (3) causal-coding style modularization that makes continual learning robust by enforcing approximate commutativity of context-conditioned updates.

The core technical insight is that it can be effective to define *weakness* to be the (local) Bayesian evidence/volume object that SLT already analyzes. Under this definition, the LLC is literally the asymptotic invariant of weakness in the large-sample (or low-temperature) limit. We then show how tropicalization (log-sum-exp dominance) yields a min-plus semiring in which MAP selection over a DAG of candidate factorizations becomes shortest-path dynamic programming. This yields a clean mathematical model of “updating many approximate factorizations per context”: quantale-message-passing over a refinement DAG.

We apply the framework to (i) stagewise degeneracy changes observed during standard backprop Transformer training, interpreted as posterior switching between neighborhoods with different LLCs, and to (ii) predictive/causal-coding Transformers, where causal gates and clarity regularizers maintain a modular factorization that stabilizes continual learning under distribution shift. Finally, we connect this to incremental compression: the refinement DAG can be viewed as a DAG of compression programs, while causal-coding NNs implement those compressors. There is also a connection to the TransWeave framework, which can be used to assist in adapting the solutions to the dynamic-programming problem related to factorization as distributions shift.

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1 Introduction

To approach AGI one must not be restricted to training models in one stationary regime. AGI models or systems must continuously adapt across shifting domains, tasks, and user intents. A useful theory should therefore describe not just a parametric model class, but a *system*: a coupling of (i) a base learner, (ii) a structure controller that maintains a modular causal/factorized world-model, and (iii) a meta-learner that keeps the whole mechanism stable under distribution shift.

The empirical clue: degeneracy changes. Recent work tracks a quantity called the "local learning constant" (LLC) during Transformer training process and finds stage boundaries at critical points of the LLC curve; these stages align with changes in internal structure and behavior. Their SLT framing uses local free energy

$$F_n(W^*) = -\log \int_{W^*} \exp(-n\ell_n(w))\phi(w) dw$$

and the asymptotic expansion

$$F_n(W^*) = n\ell_n(w^*) + \lambda(w^*) \log n + O_p(\log \log n),$$

with posterior concentration approximated by a log-sum-exp dominance over neighborhoods. This is the first pillar: *LLC as degeneracy / complexity penalty in learning dynamics*.

The conceptual gap. LLC is usually computed as a geometric invariant of a singular statistical model. Weakness, in contrast, is often introduced as an Occam-like preference for "indistinct" hypotheses, sometimes formalized using quantales with soft inf-convolution. The gap in many discussions is that weakness and LLC look related but live in different languages.

The main move. We define weakness to be exactly (local) evidence, i.e. the same integral appearing in SLT. Then LLC is not approximated by weakness: it *is* the asymptotic log-volume exponent of weakness. To study composition and control, we place this evidence object in a thermodynamic quantale whose tropical (zero-temperature) limit becomes the min-plus semiring. In that semiring, MAP over a refinement DAG of candidate factorizations is shortest-path dynamic programming.

What this buys you.

- A single mathematical object (*weakness as evidence*) whose asymptotics yields LLC, and whose algebra yields compositional control laws.

- A rigorous model of “many approximate factorizations per context” as inference over a refinement DAG.
- A clean bridge between: stagewise degeneracy changes in backprop Transformers; and the design logic of causal-coding systems that maintain modularity and resist catastrophic forgetting under shift.

2 A Systems Architecture: Base Learner + Structure Controller + Meta-Controller

2.1 The three loops (and why they are distinct)

Informal picture. We separate three time scales.

1. **Inference loop** (fast): infer latent activities given input and current parameters. In predictive coding this is a relaxation dynamics that reduces free energy.
2. **Learning loop** (medium): update neural parameters (or program library) using local errors and gates.
3. **Structure/Meta loop** (slow): update modular factorization, gating policies, clarity strengths, and shift segmentation so as to keep the system in a regime where continual-learning guarantees hold.

The thesis of this paper is that (3) can be formalized as inference and control on a refinement DAG, and that LLC/weakness provides a quantitative signal for when refactorization is needed.

2.2 Formal system model

Definition 1 (Contexts and data streams). Let $\mathcal{C}$ be a set of contexts (domains/tasks/interaction regimes). A nonstationary training process produces a time-indexed stream $(c_t, x_t)_{t \geq 1}$ with $c_t \in \mathcal{C}$ and $x_t \sim P_{c_t}$.

Definition 2 (Modular learner state). The learner state is modularized as

$$\Theta \cong \Theta^{(1)} \otimes \dots \otimes \Theta^{(K)}$$

where $\Theta^{(k)}$ is the parameter/state of module k (e.g. attention head, channel group, or program-block).

Definition 3 (Structure objects and refinement DAG). A *factorization* f is a choice of modular decomposition of the learner’s internal representation and/or parameters (not necessarily equal to the fixed physical module partition). Let $\mathcal{F}$ be a collection of candidate factorizations and let

$$\mathcal{G} = (\mathcal{F}, E)$$

be a directed acyclic graph (DAG) whose edges represent refinement moves (e.g. splitting a module, introducing a gating separation, or adding a sparse interaction edge). We call $\mathcal{G}$ the *refinement DAG*.

Why multiple factorizations? A large model can represent many incompatible but locally-useful ways of factorizing data. Different contexts may prefer different factorizations. Also, distribution shift can render a previously good factorization unstable. Hence we maintain a *distribution over* factorizations and update it online.

Definition 4 (System components). A system consists of:

- a base update rule $U_c : \Theta \rightarrow \Theta$ (neural or programmatic) for each context c ;
- a structure controller that maintains a posterior (or score function) over $f \in \mathcal{F}$;
- a meta-controller Φ that adjusts estimation/calibration/regularization/segmentation knobs so as to keep the system stable under shift. (This is where TransWeave or other sophisticated AI methodologies can help on the adaptive learning side.)

3 Singular Learning Theory: Local Evidence, Free Energy, and LLC

3.1 Why local evidence (not global risk) is the right basic object

SLT explains generalization and learning dynamics in *singular* models by tracking how much posterior mass sits near different neighborhoods of the parameter space. This is fundamentally a *volume* computation, not a Hessian computation. Local evidence captures both loss (energy) and degeneracy (entropy-like volume in parameter space). LLC appears as the exponent governing the polynomial decay of evidence with sample size.

3.2 Definitions

Definition 5 (Empirical and population losses). Fix a context $c \in \mathcal{C}$. Let $\ell_c(w)$ be the population loss (e.g. negative log-likelihood risk), and $\ell_{n,c}(w)$ the empirical loss from n samples from P_c .

Definition 6 (Local evidence and local free energy). Let $W \subset \mathbb{R}^d$ be a measurable neighborhood and $\phi(w)$ a prior density positive on W . Define the *local evidence* and *local free energy* by

$$Z_{n,c}(W) := \int_W \exp(-n \ell_{n,c}(w)) \phi(w) dw, \quad (1)$$

$$F_{n,c}(W) := -\log Z_{n,c}(W). \quad (2)$$

Definition 7 (Local learning coefficient (LLC)). Informally, the LLC $\lambda(w^*)$ is the coefficient of $\log n$ in the asymptotic expansion of $F_{n,c}(W^*)$ around a neighborhood W^* of a (population) minimizer w^* . A precise definition can be given in terms of the largest pole of the associated zeta function $\zeta(z) = \int (\ell_c(w) - \ell_c(w^*))^z \phi(w) dw$ after analytic continuation.

3.3 Watanabe’s local free energy expansion

For regular models, Laplace’s method yields $F_n \approx n\ell(\hat{w}) + \frac{d}{2} \log n$. For singular models, the “ $\frac{d}{2}$ ” becomes a rational number λ determined by the singular geometry. This λ is the LLC.

Theorem 8 (Local free energy expansion (SLT)). *Under standard SLT regularity conditions (analyticity, proper prior, etc.), for a neighborhood W^* containing a local minimum w^* of the population loss, there exist constants $\lambda(w^*) \in \mathbb{R}_{>0}$ and $m(w^*) \in \mathbb{N}$ such that*

$$F_{n,c}(W^*) = n \ell_{n,c}(w^*) + \lambda(w^*) \log n - (m(w^*) - 1) \log \log n + O_p(1).$$

Remark 9 (Stage transitions as posterior switching). When multiple neighborhoods compete, the posterior mass is approximately concentrated in the neighborhood(s) with smallest $F_{n,c}(W^*)$, by a log-sum-exp dominance principle. As n increases, the tradeoff between the $n\ell$ term and the $\lambda \log n$ term can cause abrupt switching between neighborhoods with different degeneracies.

4 Weakness as Local Evidence in a Thermodynamic Quantale

4.1 Why quantales?

A system perspective needs a compositional algebra: how do modules compose, how do alternatives compete, and how does “best explanation” emerge? Quantales provide exactly this: a lattice of alternatives (joins) together with a notion of sequential composition (monoidal product) that distributes over joins. Thermodynamic (soft-min) quantales model log-sum-exp dominance; their tropical limit models hard-min composition.

4.2 Thermodynamic semiring and quantale

A scalar warm-up. Before defining kernel-valued quantales, it helps to see the scalar case: evidence sums over alternatives and multiplies over independent compositions. Taking negative logs turns multiplication into addition and sums into log-sum-exp.

Definition 10 (Thermodynamic addition and tropical limit). Fix $\varepsilon > 0$. For $a, b \in [0, \infty]$, define

$$a \oplus_\varepsilon b := -\varepsilon \log(e^{-a/\varepsilon} + e^{-b/\varepsilon}), \quad a \otimes b := a + b,$$

with the conventions $e^{-\infty/\varepsilon} = 0$ and $-\varepsilon \log 0 = +\infty$.

Lemma 11 (Soft-min bounds and tropicalization). *For all $a, b \in [0, \infty]$,*

$$\min\{a, b\} - \varepsilon \log 2 \leq a \oplus_\varepsilon b \leq \min\{a, b\}.$$

In particular, $\lim_{\varepsilon \rightarrow 0+} (a \oplus_\varepsilon b) = \min\{a, b\}$.

Proof. Assume $a \leq b$ (the other case is symmetric). Then

$$a \oplus_\varepsilon b = -\varepsilon \log(e^{-a/\varepsilon}(1 + e^{-(b-a)/\varepsilon})) = a - \varepsilon \log(1 + e^{-(b-a)/\varepsilon}).$$

Since $\log(1 + e^{-(b-a)/\varepsilon}) \in (0, \log 2]$, the bound follows. $\square$

Proposition 12 (Semiring/quantale structure). *$([0, \infty], \oplus_\varepsilon, \otimes)$ is a commutative semiring with additive unit $+\infty$ and multiplicative unit 0. Moreover, extending $\oplus_\varepsilon$ to arbitrary (countable) joins by*

$$\bigoplus_{\varepsilon, i} a_i := -\varepsilon \log\left(\sum_i e^{-a_i/\varepsilon}\right),$$

one obtains a complete distributive structure (a commutative quantale) in the standard measurable setting. The tropical limit $\varepsilon \rightarrow 0$ yields the idempotent min-plus quantale $([0, \infty], \min, +)$.

4.3 Kernel-valued Schrödinger-bridge quantale (for compositional structure)

To model composition of multi-step mechanisms, we need costs between states, not just scalars. Thermodynamic (entropic) inf-convolution provides the correct “soft path composition”: sequential composition sums costs along a path and then aggregates over intermediate states via log-sum-exp. This is precisely the operator/heat-kernel view that underlies both optimal transport and many “weakness” formalisms.

Definition 13 (Schrodinger-bridge cost quantale). Let (X, μ) be a σ -finite measure space and let $\mathcal{Q}_{\text{SB}}(\varepsilon)$ be the set of measurable kernels $C : X \times X \rightarrow [0, \infty]$. Define, for $\varepsilon > 0$:

$$(C \oplus D)(x, z) := \min\{C(x, z), D(x, z)\}, \quad (3)$$

$$(C \otimes_{\varepsilon} D)(x, z) := -\varepsilon \log \int_X \exp\left(-\frac{C(x, y) + D(y, z)}{\varepsilon}\right) d\mu(y). \quad (4)$$

Then $(\mathcal{Q}_{\text{SB}}(\varepsilon), \oplus, \otimes_{\varepsilon})$ is a quantale; its $\varepsilon \rightarrow 0$ limit is the classical “min-plus” inf-convolution:

$$(C \otimes_0 D)(x, z) := \inf_{y \in X} (C(x, y) + D(y, z)).$$

Remark 14 (Gibbs operator view). Define the integral operator $(K_C \varphi)(x) = \int e^{-C(x, y)/\varepsilon} \varphi(y) d\mu(y)$. Then $K_{C \otimes_{\varepsilon} D} = K_C K_D$, i.e. composition of costs corresponds to composition of positive operators. This is the cleanest compositional semantics for “weakness as thermodynamic volume.”

4.4 The key identification: weakness = (local) evidence

At this point, we can make the promised identification. Evidence is a partition function for the empirical loss. Weakness should therefore be *defined* as that partition function (globally or locally), not as an external proxy such as a margin-like quantity. LLC is then the asymptotic exponent of weakness.

Definition 15 (Weakness as (local) evidence). Given context c and neighborhood $W \subset \mathbb{R}^d$, define the *weakness* of W by

$$\mathcal{W}_{n,c}(W) := Z_{n,c}(W) = \int_W \exp(-n \ell_{n,c}(w)) \phi(w) dw.$$

Equivalently, define the *weakness cost* by its negative log

$$\mathcal{E}_{n,c}(W) := -\log \mathcal{W}_{n,c}(W) = F_{n,c}(W).$$

Proposition 16 (LLC is the log-volume exponent of weakness). *Under the SLT assumptions of Theorem 1, for a neighborhood W^* around w^* ,*

$$\mathcal{W}_{n,c}(W^*) = \exp(-n \ell_{n,c}(w^*)) \cdot n^{-\lambda(w^*)} \cdot (\log n)^{m(w^*)-1} \cdot \exp(O_p(1)).$$

In particular, $\lambda(w^)$ is the asymptotic exponent governing polynomial decay in n of weakness.*

Proof. Exponentiate the expansion of $F_{n,c}(W^*) = -\log \mathcal{W}_{n,c}(W^*)$. □

5 Compositionality: How LLC Decomposes Under (Exact) Factorization

5.1 Why compositionality matters

A systems theory should say how complexity/degeneracy decomposes across modules. If a model factorizes exactly into independent pieces, the evidence factorizes: volumes multiply and free energies add. This implies that LLC should add across independent modules. This is the foundational case; later sections treat approximate factorization and sparse interactions.

Assumption 17 (Exact factorization). Suppose $w = (w^{(1)}, \dots, w^{(K)})$ and, in a neighborhood $W^* = \prod_k W_k^*$,

$$\ell_{n,c}(w) = \sum_{k=1}^K \ell_{n,c}^{(k)}(w^{(k)}), \quad \phi(w) = \prod_{k=1}^K \phi_k(w^{(k)}).$$

Proposition 18 (Evidence, free energy, and LLC add under exact factorization). *Under Assumption 1,*

$$Z_{n,c}(W^*) = \prod_{k=1}^K Z_{n,c}^{(k)}(W_k^*), \quad F_{n,c}(W^*) = \sum_{k=1}^K F_{n,c}^{(k)}(W_k^*).$$

If each factor has an SLT expansion with LLC λ_k , then the global LLC is

$$\lambda = \sum_{k=1}^K \lambda_k,$$

and similarly multiplicities add in the expected way.

Proof. Factorization of Z follows from Fubini/Tonelli and the product form of the exponent and prior. Then $F = -\log Z$ gives additivity, and the $\log n$ coefficients add termwise. $\square$

6 Refinement DAGs: Many Approximate Factorizations and Context-Dependent Selection

6.1 Why a refinement DAG captures “emergent causal/Bayesian structure”

A single static factorization is too rigid. A large learner can represent many decompositions; contexts select among them. We represent the space of decompositions as a refinement DAG: edges correspond to safer, incremental refactorizations (split a module; add a sparse interaction edge; etc.). A prior over this DAG encourages simplicity (fewer edges, sparser cliques).

6.2 Scoring a factorization by weakness/evidence

Definition 19 (Factorization-dependent evidence). Let $f \in \mathcal{F}$ index a candidate factorization. We write $w = (w_f, \eta_f)$ where w_f are base parameters and η_f are factorization-specific structural/gating parameters. Define the (local) evidence under f by

$$Z_{n,c}(f; W) := \int_W \exp(-n \ell_{n,c}(w_f, \eta_f)) \phi_f(w_f, \eta_f) dw_f d\eta_f,$$

and the free energy by $F_{n,c}(f; W) := -\log Z_{n,c}(f; W)$.

Definition 20 (Refinement-DAG prior). Let $\pi(f)$ be a prior over $\mathcal{F}$ (e.g. penalize refinement depth or edge count). Define the posterior score by

$$\mathcal{J}_{n,c}(f) := F_{n,c}(f; W_f^*) - \log \pi(f),$$

where W_f^* is a chosen neighborhood around the fitted point for factorization f .

Active factorization map. Define the MAP-selected factorization

$$f^*(c; n) \in \arg \min_{f \in \mathcal{F}} \mathcal{J}_{n,c}(f).$$

This map can be highly non-smooth as a function of c and n . The goal of causal coding (and meta-control) is to keep it *mostly stable* and to make its jump set interpretable.

6.3 Tropicalization and the “piecewise-constant map” theorem

Because $\mathcal{J}_{n,c}(f)$ is a negative log-evidence, it is a log-sum-exp object. When n is large (or temperature small), it tropicalizes: the best term dominates. Thus selection becomes min-plus algebra, and the map $c \mapsto f^*(c; n)$ becomes piecewise constant, with explicit inequality boundaries where two candidates tie.

Assumption 21 (Asymptotic expansion per factorization). Assume each factorization has an SLT-like expansion

$$F_{n,c}(f; W_f^*) = n \hat{\ell}_c(f) + \lambda_c(f) \log n + O_c(1),$$

where $\hat{\ell}_c(f)$ is the achieved empirical loss level and $\lambda_c(f)$ the effective LLC.

Proposition 22 (Jump-set characterization). *Fix n . For any two candidates f_1, f_2 , the region of contexts where f_1 is preferred over f_2 is*

$$\left\{ c \in \mathcal{C} : \mathcal{J}_{n,c}(f_1) \leq \mathcal{J}_{n,c}(f_2) \right\}.$$

Under Assumption 2, ignoring $O(1)$ terms, this inequality is approximately

$$n(\hat{\ell}_c(f_1) - \hat{\ell}_c(f_2)) + (\lambda_c(f_1) - \lambda_c(f_2)) \log n \lesssim \log \frac{\pi(f_1)}{\pi(f_2)}.$$

Hence the jump set is approximately the union of codimension-1 surfaces where the left-hand side is equal.

7 Dynamic Programming View: Refinement-DAG MAP as Shortest Path in Min-Plus

7.1 Why dynamic programming emerges

If every refinement step adds a local structural change (a “micro-compiler pass”) whose cost is additive, then the total score of a factorization is a sum over edges along a path in the refinement DAG. Minimizing a sum over paths is exactly a shortest-path problem. Over the thermodynamic (soft) semiring it becomes log-sum-exp message passing; in the tropical limit it becomes min-plus DP.

Assumption 23 (Path-decomposable factorization scores). Assume each $f \in \mathcal{F}$ corresponds to a path $p = (e_1, \dots, e_m)$ from a root f_0 in a refinement DAG, and that for each context c and sample size n ,

$$\mathcal{J}_{n,c}(f) = \mathcal{J}_{n,c}(f_0) + \sum_{i=1}^m \kappa_{n,c}(e_i),$$

where $\kappa_{n,c}(e)$ is the incremental cost of refinement edge e (including prior penalty).

Theorem 24 (Refinement-DAG MAP is shortest path (tropical limit)). *Under Assumption 3, MAP selection equals the shortest path in the min-plus semiring:*

$$f^*(c; n) \in \arg \min_{p: f_0 \rightarrow f} \sum_{e \in p} \kappa_{n,c}(e).$$

Because $\mathcal{G}$ is acyclic, the argmin can be computed exactly by dynamic programming in topological order.

Proof. In a DAG, every node’s optimal cost satisfies the Bellman recursion

$$\text{Cost}(v) = \min_{u \rightarrow v} (\text{Cost}(u) + \kappa(u \rightarrow v)),$$

which is min-plus distributivity. This yields the shortest-path cost to each node and thus the MAP node(s). $\square$

Remark 25 (Thermodynamic (soft) DP as quantale-message-passing). At finite temperature $\varepsilon > 0$, the min is replaced by $\oplus_\varepsilon$:

$$\text{Cost}_\varepsilon(v) = \bigoplus_{\varepsilon, u \rightarrow v} (\text{Cost}_\varepsilon(u) + \kappa(u \rightarrow v)),$$

which is exactly log-sum-exp message passing (soft shortest path).

8 TransWeave and Bootstrapped Multi-Stratum Transfer for Causal Factorization DAGs

8.1 Why TransWeave belongs in this story

TransWeave was originally developed as a general framework for transferring solutions between *structured* dynamic-programming (DP) or Bellman problems. Each task is specified by a state space, a Bellman operator, and a value function; transfer then amounts to learning an *intertwining map* that approximately commutes with the Bellman operators of two tasks. The same mathematical pattern appears naturally in our refinement-DAG picture:

- each distribution P or Q induces a DP problem over the refinement DAG
- the Bellman operator encodes the one-step refinement cost
- under distribution shift, we want to transfer structure from the “old” DP problem to the “new” DP problem, using a map that approximately intertwines their Bellman operators.

This is exactly the niche TransWeave is designed to fill. In particular, the “Bootstrapped Multi-Stratum TransWeave” (BMS-TransWeave) algorithm provides a way to start from a coarse, low-quality mapping between the two DP problems and iteratively improve both the mapping and the target solutions.

8.2 TransWeave in one paragraph (mathematical sketch)

At a high level, TransWeave treats each task T as a Bellman system $(\mathcal{X}_T, B_T)$:

- $\mathcal{X}_T$ is a space of value functions or policies (depending on the formulation);
- $B_T : \mathcal{X}_T \rightarrow \mathcal{X}_T$ is a Bellman operator, e.g. $(B_TV)(s) = \min_a \{c_T(s, a) + \gamma V(s')\}$ or a min-plus convolution on a graph.

A *transfer map* between two tasks T_1 and T_2 is an operator

$$\mathcal{T} : \mathcal{X}_{T_1} \rightarrow \mathcal{X}_{T_2}$$

that approximately satisfies a Bellman–Darboux (BD) intertwining condition

$$\mathcal{T} \circ B_{T_1} \approx B_{T_2} \circ \mathcal{T}. \quad (5)$$

Ideally, (5) ensures that if V_1^* is a (near-)fixed point of B_{T_1} , then $\mathcal{T}V_1^*$ is a good approximate fixed point of B_{T_2} . TransWeave algorithms construct $\mathcal{T}$ by minimizing a BD loss derived from (5), optionally combined with direct supervision on pairs of known solutions.

8.3 Refinement-DAG dynamic programming as a TransWeave task

In our refinement-DAG framework, each distribution over contexts induces a DP problem over the DAG:

- nodes $f \in \mathcal{F}$ are candidate factorizations;
- directed edges $e : f \rightarrow f'$ are refinement moves (splitting modules, adding or removing sparse interaction edges, changing gating structures);
- costs on edges encode the incremental change in the SLT-aligned score $S_n(c, f) = nL_{c,f}^* + \lambda_{c,f} \log n + \kappa(f)$;
- the Bellman operator B is the min-plus DP update that propagates these costs forward along the DAG .

For a given distribution P over contexts, let $(\mathcal{F}, B_P)$ be the corresponding DP problem, and $V_P(f)$ its (approximate) value function over factorizations. For a shifted distribution Q , let $(\mathcal{F}, B_Q)$ and $V_Q(f)$ be defined analogously, with task-dependent costs reflecting $L_{c,f}^*$ and $\lambda_{c,f}$ evaluated under Q rather than P .

Under this identification, TransWeave’s BD map becomes a structure-transfer operator

$$\mathcal{T} : \mathcal{X}_P \rightarrow \mathcal{X}_Q$$

between value-function spaces over the same refinement DAG, satisfying

$$\mathcal{T} \circ B_P \approx B_Q \circ \mathcal{T}.$$

8.4 Bootstrapped Multi-Stratum TransWeave for Causal Factorization DAGs (BMSTW-CF)

Informal goal. Under distribution shift from P to Q , we often have:

- *high-quality* refinement policies and value functions for $(\mathcal{F}, B_P)$,
- only *low-quality* or partially-trained policies/value functions for $(\mathcal{F}, B_Q)$.

The goal of BMSTW-CF is to iteratively improve both the transfer map $\mathcal{T}$ and the target-side value function V_Q by:

1. bootstrapping from low-stratum (weak) solutions on Q and better solutions on P ,
2. learning a BD map that fits these strata,
3. using this map to propose better initializations for Q and refine them with DP,
4. and repeating across strata until both $\mathcal{T}$ and V_Q are of high quality.

Strata over the refinement DAG. We stratify factorization nodes $f \in \mathcal{F}$ by *quality* under the new distribution Q . For instance, define a scalar score

$$q_Q(f) := -S_n(Q, f)$$

and partition $\mathcal{F}$ into strata such as “barely acceptable”, “OK”, and “good” based on quantiles of $q_Q(f)$. Early in training on Q , most nodes are in the “barely acceptable” or unknown strata.

Initial approximate transfer map. Causal coding plus meta-control already provide a rough structural alignment between P and Q :

- modules that retain high do-influence under Q are likely to correspond to modules that were important under P ;
- mixed-Hessian/commutator surrogates identify interaction edges that survive vs those that vanish under Q .

We use this information to define an initial coarse map

$$T_0 : \mathcal{F} \rightarrow \mathcal{F},$$

for example by “copying” a factorization f from P to its closest structural analog under Q (same module partitions, edges, and gates, where possible) and adjusting only obviously broken connections.

Bootstrapped BD-map learning. BMSTW-CF proceeds in strata:

1. *Low-stratum phase.*
 - Select a set of low-quality factorizations $f \in \mathcal{F}$ under Q and their parents/siblings.
 - For each such f , use T_0 to define a source-side partner f_P that is good under P (value $V_P(f_P)$ known).

- Fit a preliminary map $\mathcal{T}_1$ by minimizing a BD loss

$$\|\mathcal{T}_1(B_P V_P) - B_Q(\mathcal{T}_1 V_P)\|$$

plus a supervised term encouraging $\mathcal{T}_1(V_P)(f_P) \approx V_Q^{\text{approx}}(f)$.

2. Transfer and refinement.

- Use $\mathcal{T}_1$ to transfer better source-side value estimates V_P down to Q : for higher-quality f_P , propose initial $V_Q^{\text{seed}}(f)$ as $\mathcal{T}_1(V_P)(f_P)$.
- Run a few DP refinement passes using B_Q to locally improve V_Q near these seeds.

3. Re-stratify and iterate.

- Recompute $q_Q(f)$ and update strata.
- Restrict attention to higher strata, learn a refined $\mathcal{T}_2$ with stricter BD error tolerance.
- Repeat the transfer-and-refinement step across strata until convergence.

We call this overall procedure *Bootstrapped Multi-Stratum TransWeave for Causal Factorization DAGs* (BMSTW-CF).

8.5 Role of BMSTW-CF in handling distribution shifts

Upgrading a rough mapping to a high-quality one. Above we have argued that causal coding plus meta-learning naturally produce a coarse, evolving picture of which factorizations in the refinement DAG remain good under shift (small leakage, small commutators, small interaction LLC corrections $\Delta\lambda$). However, this picture is noisy and local. BMSTW-CF offers a principled way to turn that coarse picture into a high-quality mapping and refined target-side policies:

- treat the DP problems induced by P and Q as Bellman systems $(\mathcal{F}, B_P)$ and $(\mathcal{F}, B_Q)$;
- treat the causal-coding-induced correspondence as an initialization T_0 between their value-function spaces;
- use BMS-TransWeave to learn an intertwiner $\mathcal{T}$ satisfying $\mathcal{T} \circ B_P \approx B_Q \circ \mathcal{T}$ while bootstrapping better value functions on Q ;
- thereby systematically upgrade the rough initial mapping into a better one that respects the DP structure and the refined SLT/weakness scores.

Architectural placement. From the systems viewpoint we’ve outlined, BMSTW-CF lives in the *slow structure/metacontrol loop*:

- the fast neural loop (in JAX or similar) performs predictive/causal coding inside whichever factorization node is currently active;
- the medium loop updates parameters and local gates based on context-specific errors;
- the slow loop, implemented in a symbolic graph engine (e.g. MeTTa/MORK), maintains the refinement DAG, runs DP over it, and uses BMSTW-CF to transfer and upgrade structures as the distribution drifts.

This allows the system to *reconcile* the causal-coding local view (supports, leakage, commutators) with the SLT/global view (LLC, weakness) in a computationally tractable way, across multiple distributions and over time.

9 Causal Coding as Maintaining a Refinement DAG in a Continual-Learning Regime

9.1 Why causal coding belongs in the structure controller

Causal coding (CC) introduces two families of mechanisms: (i) causal gates that suppress updates on edges with low do-influence, and (ii) clarity/pruning terms that remove redundant paths. From a systems viewpoint, these are not just local heuristics; they are the operations that *realize refinement steps* in the refinement DAG, and keep the model near a factorization state where context-conditioned updates commute (or almost commute).

9.2 Abstract commutator control (CCL viewpoint)

Reminder of the commutator goal. The continual-learning theorems in causal-coding work can be read as: *If updates are local to supports (small leakage) and cross-support interactions are sparse/small, then sequential training approximates joint training and catastrophic forgetting is bounded.*

Definition 26 (Approximate locality and commutator defects (schematic)). Fix a metric/enrichment d_Θ on endomorphisms of Θ . For each context c , let $U_c : \Theta \rightarrow \Theta$ be the realized update and V_c an ideal support-local update. Define locality error $\varepsilon_c := d_\Theta(U_c, V_c)$. For a pair (c, c') , define commutator defect $\Delta_{c,c'} := d_\Theta(V_c \circ V_{c'}, V_{c'} \circ V_c)$.

Why this matters for the refinement DAG. Factorization states f determine support sets and routing paths. Good factorizations make most pairs (c, c') have small overlap and thus small commutators. Distribution shifts threaten this by changing which modules matter, or by corrupting influence estimation.

9.2.1 Meta-objectives for staying in the small-commutator, sufficient-LLC regime under shift

The commutator calculus says: if context-conditioned updates are (approximately) local to causal modules, and if the remaining cross-context interactions are sparse, then continual learning is stable and path dependence is controlled by a *sparse confusion graph*. In practice, the difficult part is not the algebra downstream of these assumptions, but maintaining the assumptions online under distribution shift: influence estimates drift, gates flip, leakage cascades across depth, and the confusion graph can densify. A principled cure is to make the relevant knobs *meta-learned* with an explicit objective that punishes the very quantities that break the theorems: locality error and commutator/confusion defects. This is exactly the move advocated in the distribution-shift addendum, which proposes learning calibration/thresholds/clarity/segmentation policies so as to keep ε_c and $\Delta_{c,c'}$ small for most contexts/pairs while controlling compute. [?] [?]

Mathematical setting (recap). We work with a modularized learner state $\Theta \simeq \Theta_1 \otimes \cdots \otimes \Theta_K$ and a family of context-conditioned update operators $U_c : \Theta \rightarrow \Theta$. The approximate categorical causal-continual-learning (CCL) viewpoint treats continual-learning stability as a problem of controlling two types of errors: (i) per-context locality error $\varepsilon_c := d_\Theta(U_c, V_c)$ to an ideal support-local update V_c , and (ii) per-pair commutator defect $\Delta_{c,c'} := d_\Theta(V_c \circ V_{c'}, V_{c'} \circ V_c)$. These combine as the per-pair confusion cost

$$\delta_{c,c'} := 2(\varepsilon_c + \varepsilon_{c'}) + \Delta_{c,c'}. \quad (6)$$

When the set of “confusing pairs” is sparse, global path dependence scales (essentially) linearly in $\sum_{\{i,j\} \in E_{\text{conf}}(\Sigma)} \delta_{c_i, c_j}$ rather than quadratically in the number of contexts. [?]

What to meta-optimize (informal). A meta-controller should be rewarded for (a) good current performance, (b) low forgetting on a small anchor set, (c) keeping updates local (low leakage outside estimated supports), and (d) keeping commutator/confusion small on sampled context pairs, all under a compute/latency budget. The shift note spells this out as a direct meta-objective connected to $(\varepsilon_c, \Delta_{c, c'})$ and the confusion graph. [?]

Meta-parameterization. Let ω denote “control” parameters used by causal coding / predictive-coding (PC) implementations, e.g.:

- gating thresholds τ_c ;
- uncertainty radii / calibration factors $\beta_{c,k}$ for per-module influence estimates;
- clarity/anti-overlap regularization strengths;
- exploration rates (to avoid frozen-gate failure modes);
- segmentation / change-point thresholds controlling context prototypes and module allocation.

This matches the architectural sketch in which a meta-controller learns calibration, confidence-aware gating, clarity strengths, and shift-detection/segmentation policies. [?]

Observed proxies for the theorem quantities. Because V_c and $\Delta_{c, c'}$ are idealized, we use measurable proxies.

(1) *Leakage proxy for locality error.* Given an estimated support $\hat{S}_c$ (from confidence-aware gating), define a measured leakage proxy $\hat{\varepsilon}_c$ as the magnitude of update outside $\hat{S}_c$; e.g. in a continuous neural setting,

$$\hat{\varepsilon}_c := \sum_{k \notin \hat{S}_c} \|\Delta\theta^{(k)}\|, \quad (7)$$

as suggested in the stability-map discussion. [?]

(2) *Commutator/confusion proxy.* Exact computation of $d_{\Theta}(U_c \circ U_{c'}, U_{c'} \circ U_c)$ is expensive, so we estimate it sparsely and/or via cheap proxies. In smooth neural systems one proxy is the first-order commutator approximation based on gradients and Hessian-vector products,

$$\hat{\Delta}_{c, c'} \approx \left\| \nabla^2 L_c(\theta) \nabla L_{c'}(\theta) - \nabla^2 L_{c'}(\theta) \nabla L_c(\theta) \right\|, \quad (8)$$

or a blockwise proxy based on cross-module gradient/Hessian overlaps. [?]

(3) *Confusion-cost proxy.* Define

$$\hat{\delta}_{c, c'} := 2(\hat{\varepsilon}_c + \hat{\varepsilon}_{c'}) + \hat{\Delta}_{c, c'}. \quad (9)$$

This mirrors the note’s schematic $\hat{\delta}_{c, c'} := 2(\hat{\varepsilon}_c + \hat{\varepsilon}_{c'}) + \hat{\Delta}_{c, c'}$ and supports maintaining an approximate confusion graph by sampled estimates. [?]

From estimation error to locality error: the stability map. Under distribution shift, the upstream failure mode is misestimation of influence/support, producing leakage that cascades. The shift note proposes explicitly learning a *stability map* from influence estimation uncertainty (and drift features, thresholds, compute budget) to observed leakage $\hat{\epsilon}_c$, and using this map to tune ω to keep $\hat{\epsilon}_c$ below target. [?]

Formally, let an influence estimator return for each module k a mean/uncertainty $(\mu_{c,k}, \sigma_{c,k})$ and let confidence-aware gating form supports from lower confidence bounds (LCBs), e.g.

$$\text{LCB}_{c,k} := \mu_{c,k} - \beta_{c,k}, \quad \hat{S}_c := \{k : \text{LCB}_{c,k} \geq \tau_c\}, \quad (10)$$

with optional exploratory mixing to avoid frozen gates. [?] A (learned) stability map is then a function S_ψ such that, operationally,

$$\hat{\epsilon}_c \approx S_\psi(\{\sigma_{c,k}\}_{k=1}^K, \tau_c, \text{drift stats, compute budget}), \quad (11)$$

and the meta-controller uses S_ψ to adjust $(\tau_c, \beta_{c,k})$ to keep $\hat{\epsilon}_c$ small. [?]

A concrete meta-objective (rigorous statement). Let $\mathcal{A}$ be an “anchor” distribution over past contexts used to monitor forgetting, and let $\mathcal{P}$ be a sampling scheme over context pairs used to estimate confusion/commutator proxies. Define the meta-objective

$$J(\omega) := \mathbb{E}[\text{CurrLoss}] + \alpha \mathbb{E}_{c \sim \mathcal{A}}[\text{AnchorForget}(c)] + \beta \mathbb{E}_{(c,c') \sim \mathcal{P}}[\hat{\delta}_{c,c'}] + \gamma \mathbb{E}[\text{ExtraCompute}], \quad (12)$$

where $\hat{\delta}_{c,c'}$ is as above. This is the explicit schematic objective proposed in the shift note. [?]

Interpretation: “stay in the theorem regime”. Minimizing (12) targets precisely the premises that make the approximate categorical theorem useful: (i) small per-context locality error (via leakage penalties and stability-map tuning), and (ii) small commutator/confusion on most pairs (via sampled commutator proxies and confusion-graph maintenance), with explicit compute awareness. [?] This turns the CCL theorem from a post-hoc analysis tool into a *control principle* for a continual learner.

Connection to mixed-Hessian / commutator surrogates (why this is computable). In the smooth setting, commutators of context gradient flows can be bounded in terms of mixed Hessian blocks and (out-of-support) gradients. Concretely, causal modularity is defined by small residual gradient outside support and small cross-Hessian blocks between support and non-support modules; these quantities control contributions to the Lie bracket (commutator) of gradient vector fields. [?] This justifies using mixed-Hessian/HVP proxies (and blockwise overlaps) as inexpensive estimates of $\hat{\Delta}_{c,c'}$ inside (12). [?]

Where this meets “AGI cognitive architecture”. The architectural lesson is that “causal coding” is not only a base-learning rule (gated local updates); it is also a *self-regulation problem* over nonstationary environments. The meta-controller is responsible for:

1. maintaining calibrated uncertainty and confidence-aware gating (to prevent cascades),
2. allocating exploration/probes to avoid frozen gates and discover re-emergent relevance,
3. detecting mechanism shifts and re-segmenting contexts / expanding modules,

4. keeping the confusion graph sparse by directing replay and clarity increases to persistent edges.

These design requirements are explicitly motivated as the practical route for robust causal estimation under shift and for preserving the linear-in-confusing-pairs regime. [?]

Remark (how this interfaces with SLT/LLC in the broader paper). In the main text, you can now identify an SLT-aligned “weakness/evidence” quantity (whose tropical asymptotics yield LLC) as an additional term in the control objective: the controller should maintain not only low commutator/leakage but also sufficient local evidence volume (to avoid collapse into an over-constrained, brittle subspace). This subsection is the right place to introduce that additional regularizer once your paper has defined weakness in SLT terms.

9.3 Distribution shift and meta-learning as “keeping the premise errors small”

Under shift, influence estimates can become wrong and gating can leak updates into non-support modules. In deep networks, small leakage can cascade, densifying the confusion graph and violating the premises needed for commutator control. This motivates a meta-controller that learns calibration, thresholds, and segmentation to maintain small locality errors and small commutator defects with high probability.

Definition 27 (Stability map (schematic)). Let $I(c, k)$ be the true influence profile and $\hat{I}(c, k)$ an online estimate. Assume a gate-stability condition for the update map:

$$d_{\Theta}(U_c(\hat{I}), U_c(I)) \leq L_{\text{gate}} \|\hat{I} - I\| + \varepsilon_{\text{alg}}.$$

Then estimation error propagates to locality error ε_c .

10 Transformers as a Key Application

10.1 Backprop Transformers: degeneracy stages as posterior switching

In standard backprop training, we can still interpret SGD as a noisy optimizer that explores regions of the loss surface. The degeneracy-stage picture says: as training progresses and effective sample size grows, the system may switch between “modes” (neighborhoods) with different loss and different LLC, because the local free energy tradeoff changes. This is exactly a log-sum-exp / tropicalization story: posterior mass concentrates at the best free-energy region.

Definition 28 (Neighborhood competition model). Let $\{W_i^*\}_{i \in I}$ be candidate neighborhoods around minima (or effective minima). Define $F_n(W_i^*)$ and say the active neighborhood is

$$i^*(n) \in \arg \min_i F_n(W_i^*).$$

Remark 29 (Why stage boundaries occur). If $F_n(W_i^*) = n\ell_i + \lambda_i \log n + O(1)$, then differences

$$F_n(W_2^*) - F_n(W_1^*) = n(\ell_2 - \ell_1) + (\lambda_2 - \lambda_1) \log n + O(1)$$

can change sign at a critical n , causing abrupt switches. This is the mathematically clean version of “developmental stage transitions” in the degeneracy-based picture.

10.2 PC/CC Transformers: modules, gates, and clarity penalties

A Transformer already has natural modularity: attention heads and MLP channels. A predictive-coding Transformer adds activity variables and local error signals. Causal coding then uses do-influence to gate which heads/channels update for a given context, and uses clarity penalties to prune redundant routes. In our systems view, these mechanisms implement refinement moves (splits/pruning/routing changes) that update the refinement DAG and keep commutators small.

Definition 30 (Transformer module index set). Let modules be indexed by $k \in \{1, \dots, K\}$, where a module may be an attention head (ℓ, h) , an MLP channel group (ℓ, j) , etc. A factorization f can group these modules differently (e.g. by task-routing partitions).

Weakness/LLC signal for “useful degeneracy”. Over-pruning can collapse the effective parameter space into a too-skinny subspace (loss of adaptive slack); under-pruning yields dense coupling and catastrophic interference. This paper’s quantitative proposal is: track module-wise and factorization-wise weakness (evidence) and LLC, and meta-control the clarity/gating to maintain a target regime:

- most contexts have small supports and small leakage ε_c ;
- most context pairs have small commutator defects $\Delta_{c,c'}$;
- the active factorization has enough degeneracy (LLC) to adapt under shifts, without collapsing.

11 Incremental Compression View: Learning the Refinement DAG as Incremental Structure Learning

11.1 Why incremental compression helps

Neural nets are one kind of compressor. Incremental compression offers a complementary lens: a learner maintains a library of reusable “features” (programs) organized into modules. Contexts have supports defined by compression gaps under interventions (disable a module). This is exactly the causal-coding support notion, but in bits rather than gradients. The refinement DAG then becomes a DAG of compression-program refinements.

Definition 31 (Incremental decomposition and code length). Given an input string x of length N , an incremental decomposition is a sequence of features $F = (f_1, \dots, f_s)$ and residual r_s with two-part code length

$$L_s(F; x) := \sum_{i=1}^s K(f_i) + K(r_s).$$

A global library $\Theta = (F^{(1)}, \dots, F^{(K)})$ stores features in modules.

Definition 32 (Do-influence as compression gap and context support). For a context c , let $L_c(\Theta)$ be expected code length under an incremental compressor that uses Θ . Let $\Theta \setminus k$ be Θ with module k disabled. Define

$$I(c, k; \Theta) := L_c(\Theta \setminus k) - L_c(\Theta), \quad \mathcal{S}_c(\Theta) := \{k : I(c, k; \Theta) \geq \tau\}.$$

Connection to refinement DAG. A refinement edge corresponds to adding/splitting a feature module, or changing reuse policies. The refinement-DAG DP decides which library structure is best for the current stream segment.

12 Computational Scaling and an Implementation Sketch (Neural + Symbolic Kernel)

12.1 How big are refinement DAGs likely to be?

The space of all partitions/graphs over K modules is enormous (super-exponential). A workable system must represent only a sparse, structured subset: hierarchical refinements, sparse interaction edges, and per-context gating patterns. In that regime, the refinement DAG can be polynomial in K .

A simple scaling model. Let K be the number of physical modules we expose (heads, channel groups, etc.). Assume: (i) each module participates in at most d interaction edges in the sparse interaction graph; (ii) refinements are hierarchical splits with branching factor at most b ; (iii) we keep at most M candidate factorizations per “region” of context space. Then nodes scale like $|\mathcal{F}| \approx O(MK)$ to $O(MK \log K)$ in typical hierarchical schemes, and edges $|E| \approx O(|\mathcal{F}|b)$.

Toy to frontier-scale orders of magnitude. For toy Transformers, K can be $\sim 10^2$; keeping $M \sim 10^2$ candidates gives $|\mathcal{F}| \sim 10^4$. For medium systems, $K \sim 10^3$ and $M \sim 10^3$ yields $|\mathcal{F}| \sim 10^6$. For frontier-scale systems, K depends on granularity (heads vs head-groups vs block-level); coarser granularity keeps K in the low thousands, making $|\mathcal{F}|$ in the 10^6 - 10^7 range plausible.

12.2 Why split implementation (e.g. JAX for NN, MeTTa/MORK for DAG control)?

The refinement DAG controller is symbolic/graphical: dynamic programming, min-plus message passing, discrete segmentation, and constraint propagation. This is naturally expressed in a reduction-kernel / logic substrate. The base learner is differentiable and batched: natural for JAX/accelerators. Hence a split architecture:

- JAX: PC inference, gated weight updates, influence estimation (or its surrogates), LLC estimators.
- MeTTa/MORK: maintain refinement DAG, run tropical/thermodynamic DP, manage priors and segmentation policies, store and update the sparse confusion graph.

13 Discussion and Open Problems

What remains to be proven (beyond this draft).

- A fully explicit perturbation theory for how LLC changes under approximate factorization, with computable bounds in terms of mixed Hessian blocks / commutator surrogates.
- Empirical protocols for estimating module-wise weakness and LLC online with acceptable overhead.
- Design of meta-objectives that directly optimize “stay in the small-commutator, sufficient-LLC regime” under realistic shift patterns.

Why this should matter. The governing message is: treat “structure” (factorizations, modularity, pruning) as a first-class inferential object, not as a static inductive bias. Weakness-as-evidence gives a quantitative, compositional signal for structure control, and tropicalization provides the correct algebra in which that control becomes dynamic programming.

References

- [1] J. Hoogland, G. Wang, M. Farrugia-Roberts, L. Carroll, S. Wei, D. Murfet. Loss Landscape Degeneracy and Stagewise Development in Transformers. arXiv:2402.02364v3, 2025.
- [2] B. Goertzel. Causal Coding and the General Causal-Continual-Learning Theorem. 2025.
- [3] B. Goertzel. Brief Note: Handling Distribution Shifts in Causal Coding via Meta-Learning. 2025.
- [4] B. Goertzel. Quantale Weakness: A Unifying Occam-Like Bias for AGI. AGI-25 keynote slides, 2025.
- [5] S. Watanabe. Mathematical Theory of Bayesian Statistics. CRC Press, 2018.

A Perturbation theory for the local learning coefficient under approximate factorization

A.1 Why LLC is the right invariant to perturb

Singular learning theory (SLT) studies the local asymptotics of Bayesian evidence integrals. Given a nonnegative analytic *excess loss* (or energy)

$$K(\theta) = L(\theta) - L(\theta^*) \geq 0,$$

a prior density $\varphi(\theta)$, and a neighborhood U of the minimizing set, the *local evidence* is

$$Z_n(U) := \int_U \exp(-nK(\theta)) \varphi(\theta) d\theta.$$

Under standard SLT hypotheses (analyticity; φ smooth and strictly positive on U), there exist numbers $\lambda > 0$ and $m \in \mathbb{N}$ such that

$$-\log Z_n(U) = \lambda \log n - (m-1) \log \log n + O(1) \quad (n \rightarrow \infty). \quad (13)$$

We call λ the *local learning coefficient* (LLC), i.e. the local RLCT.

A convenient equivalent characterization uses the *sublevel-set volume*

$$V_K(\varepsilon) := \int_{U \cap \{K(\theta) \leq \varepsilon\}} \varphi(\theta) d\theta.$$

Then, for some continuous $v(\varepsilon)$ with $0 < v(0) < \infty$,

$$V_K(\varepsilon) = \varepsilon^\lambda (\log(1/\varepsilon))^{m-1} v(\varepsilon) \quad (\varepsilon \downarrow 0). \quad (14)$$

Thus λ is the sharp exponent controlling local posterior volume near minima.

A.2 Approximate factorization as a controlled interaction remainder

To talk about “module-wise factorization” we split parameters into blocks and isolate an *interaction remainder* R measuring coupling between blocks. The main technical fact we exploit is this: *if R is dominated by the factorized baseline, then the LLC is unchanged*. This makes LLC perturbation theory largely a problem of bounding R .

Setup . For clarity start with a two-block split $\theta = (u, v)$ with $u \in \mathbb{R}^{d_u}$, $v \in \mathbb{R}^{d_v}$, and assume K is C^2 in a neighborhood of $(0, 0)$ after translating a reference minimizer to the origin (so $K(0, 0) = 0$). Define one-block “axis restrictions”

$$K_u(u) := K(u, 0), \quad K_v(v) := K(0, v),$$

the *factorized baseline*

$$K_0(u, v) := K_u(u) + K_v(v),$$

and the *interaction remainder*

$$R(u, v) := K(u, v) - K_u(u) - K_v(v). \quad (15)$$

By construction $R(u, 0) = R(0, v) = 0$.

A.3 Exact identity: the interaction remainder is a mixed-Hessian integral

In causal-coding work, “residual couplings” are measured by mixed Hessian blocks. Here is the bridge: the interaction remainder R is *exactly* an integral of those mixed blocks, so any computable bound on mixed blocks yields a computable bound on R .

Exact identity . Let K be C^2 near $(0, 0)$. Then for all (u, v) close enough to 0,

$$R(u, v) = \int_0^1 \int_0^1 u^\top (\nabla_{uv}^2 K)(su, tv) v \, dt \, ds. \quad (16)$$

Proof. Define $F(s, t) := K(su, tv)$. Then

$$R(u, v) = F(1, 1) - F(1, 0) - F(0, 1) = \int_0^1 (\partial_t F(1, t) - \partial_t F(0, t)) \, dt.$$

By the fundamental theorem of calculus in s , $\partial_t F(1, t) - \partial_t F(0, t) = \int_0^1 \partial_s \partial_t F(s, t) \, ds$. A chain-rule computation gives $\partial_s \partial_t F(s, t) = u^\top (\nabla_{uv}^2 K)(su, tv) v$, yielding (16). $\square$

Norm bound. Let $\|\cdot\|$ be the Euclidean norm and $\|\cdot\|_{\text{op}}$ the induced operator norm. If $\sup_{(su, tv) \in U} \|\nabla_{uv}^2 K(su, tv)\|_{\text{op}} \leq \rho$ on a neighborhood U , then (16) implies

$$|R(u, v)| \leq \rho \|u\| \|v\| \leq \frac{\rho}{2} (\|u\|^2 + \|v\|^2). \quad (17)$$

A.4 Core domination lemma: when R is dominated, LLC does not change

SLT’s λ is an asymptotic/scale invariant. Multiplying K by a constant or perturbing it by a term that is everywhere a small fraction of K cannot change the exponent in (14). The following lemma formalizes this “bi-Lipschitz equivalence”.

Lemma A.1 (multiplicative equivalence preserves LLC). Let K and $\widetilde{K}$ be nonnegative functions on U and suppose there exist constants $0 < c_1 \leq c_2 < \infty$ and a neighborhood $U_0 \subseteq U$ of the minimizing set such that

$$c_1 K(\theta) \leq \widetilde{K}(\theta) \leq c_2 K(\theta) \quad \text{for all } \theta \in U_0. \quad (18)$$

Then K and $\widetilde{K}$ have the same LLC λ and the same multiplicity m in the asymptotics (13)-(14).

Proof. From (18),

$$\{\theta : K(\theta) \leq \varepsilon/c_2\} \cap U_0 \subseteq \{\theta : \widetilde{K}(\theta) \leq \varepsilon\} \cap U_0 \subseteq \{\theta : K(\theta) \leq \varepsilon/c_1\} \cap U_0.$$

Integrating φ over these sets gives the corresponding sandwich for V_K and $V_{\widetilde{K}}$. A constant rescaling of ε does not change the leading $\varepsilon^\lambda (\log(1/\varepsilon))^{m-1}$ exponents in (14), hence λ, m coincide. $\square$

Corollary A.2 (dominated interaction implies $\Delta\lambda = 0$). If $K = K_0 + R$ with $K_0 \geq 0$ and there exists $\eta \in [0, 1)$ and a neighborhood U_0 such that

$$|R(\theta)| \leq \eta K_0(\theta) \quad (\theta \in U_0), \quad (19)$$

then K and K_0 have the same LLC. In particular, the “interaction LLC” $\Delta\lambda := \lambda(K) - \lambda(K_0)$ equals 0.

Proof. (19) implies $(1 - \eta)K_0 \leq K \leq (1 + \eta)K_0$ on U_0 , so Lemma A.1 applies. $\square$

A.5 A computable sufficient condition using mixed Hessian blocks (quadratic regime)

In the “quadratic regime” (each module has nondegenerate local curvature), the domination condition (19) can be certified purely from *within-block curvature lower bounds* and *mixed-block Hessian upper bounds*. Both are standard, computable quantities.

Assumption (local quadratic lower bounds). Assume there exist $\mu_u, \mu_v > 0$ and a neighborhood U_0 such that

$$K_u(u) \geq \frac{\mu_u}{2} \|u\|^2, \quad K_v(v) \geq \frac{\mu_v}{2} \|v\|^2 \quad \text{for all } (u, v) \in U_0. \quad (20)$$

Let $\mu := \min(\mu_u, \mu_v)$.

Proposition A.3 (mixed-Hessian domination test). Suppose K is C^2 on U_0 and

$$\sup_{(u,v) \in U_0} \|\nabla_{uv}^2 K(u, v)\|_{\text{op}} \leq \rho. \quad (21)$$

Then for all $(u, v) \in U_0$,

$$|R(u, v)| \leq \frac{\rho}{\mu} K_0(u, v). \quad (22)$$

Consequently, if $\rho < \mu$ then $\Delta\lambda = 0$.

Proof. By (17), $|R(u, v)| \leq \frac{\rho}{2} (\|u\|^2 + \|v\|^2)$. By (20), $K_0(u, v) = K_u(u) + K_v(v) \geq \frac{\mu}{2} (\|u\|^2 + \|v\|^2)$. Combining yields (22). If $\rho < \mu$ then (19) holds with $\eta = \rho/\mu < 1$, so Corollary A.2 gives $\Delta\lambda = 0$. $\square$

Multi-module extension (pairwise couplings). Let $\theta = (\theta^{(1)}, \dots, \theta^{(M)})$ and suppose we have a baseline $K_0(\theta) = \sum_{m=1}^M K_m(\theta^{(m)})$ satisfying $K_m(\theta^{(m)}) \geq \frac{\mu_m}{2} \|\theta^{(m)}\|^2$ locally. Assume the interaction remainder admits the bound

$$|R(\theta)| \leq \sum_{1 \leq m < n \leq M} \rho_{mn} \|\theta^{(m)}\| \|\theta^{(n)}\|.$$

Then by $ab \leq (a^2 + b^2)/2$,

$$|R(\theta)| \leq \frac{1}{2} \sum_{m=1}^M \left(\sum_{n \neq m} \rho_{mn} \right) \|\theta^{(m)}\|^2.$$

Hence if there exists $\eta \in [0, 1)$ such that for all m ,

$$\sum_{n \neq m} \rho_{mn} \leq \eta \mu_m, \quad (23)$$

then $|R(\theta)| \leq \eta K_0(\theta)$ locally and therefore $\Delta\lambda = 0$.

A.6 Beyond quadratic: order-of-vanishing (Newton) criterion

If some module directions are flat (singular), “small mixed Hessian” is not enough: a lower-order cross term can dominate at sufficiently small scale even with tiny coefficient. In that setting, what matters is the *order of vanishing* of the interaction term relative to the modules’ own leading orders. This is exactly the Newton-polyhedron logic behind SLT.

Lemma A.4 (weighted Young domination). Fix exponents $p, q > 1$ and $a, b > 0$. If $a/p + b/q > 1$, then for every $\eta > 0$ there exists $\delta > 0$ such that

$$x^a y^b \leq \eta (x^p + y^q) \quad \text{for all } x, y \in [0, \delta]. \quad (24)$$

If $a/p + b/q = 1$, then the global inequality holds:

$$x^a y^b \leq \frac{a}{p} x^p + \frac{b}{q} y^q \quad \text{for all } x, y \geq 0. \quad (25)$$

Proof sketch. For $a/p + b/q = 1$, (25) is the weighted AM-GM inequality. For $a/p + b/q > 1$, write $a/p = \alpha$, $b/q = \beta$ so $\alpha + \beta = 1 + \gamma$ with $\gamma > 0$. Then $x^a y^b = (x^{\alpha p} y^{\beta q}) x^{\gamma p} y^0$ up to relabeling, and the extra factor $x^{\gamma p}$ (or $y^{\gamma q}$) makes the bound arbitrarily small on a small neighborhood; a standard compactness argument yields (24). $\square$

Proposition A.5 (singular-safe interaction criterion). Assume there exist constants $c_u, c_v, C > 0$ and exponents $p, q > 1$, $a, b > 0$ such that locally

$$K_u(u) \geq c_u \|u\|^p, \quad K_v(v) \geq c_v \|v\|^q, \quad |R(u, v)| \leq C \|u\|^a \|v\|^b.$$

If $a/p + b/q > 1$, then $|R(u, v)| \leq \eta K_0(u, v)$ on a sufficiently small neighborhood for any chosen $\eta \in (0, 1)$, hence $\Delta\lambda = 0$.

Proof. Apply Lemma A.4 to $x = \|u\|$, $y = \|v\|$ and absorb constants c_u, c_v, C by shrinking the neighborhood. Then (19) holds and Corollary A.2 applies. $\square$

A.7 Connection to commutator surrogates (causal coding) and what is computable

Causal-coding theory controls *mixed Hessian blocks* and *out-of-support gradients* to make context-gradient flows nearly commuting. Those same mixed blocks are exactly what enters the interaction remainder bound above. Thus the same quantities that certify “small commutators” also certify “stable LLC decomposition” when within-module curvature dominates.

Link to Lie-bracket / commutator surrogates . For gradient vector fields $X_t = -\nabla L_t$, the Lie bracket can be written as

$$[X_i, X_j](\theta) = -\nabla^2 L_j(\theta) \nabla L_i(\theta) + \nabla^2 L_i(\theta) \nabla L_j(\theta),$$

and in block coordinates $\theta = (u, v)$ this contains terms in which mixed Hessian blocks $H_{uv}^{(t)}$ (and $H_{vu}^{(t)}$) are multiplied by gradient blocks. A concrete bound used in the CCL analysis is: if $\|H_{uv}^{(t)}\|, \|H_{vu}^{(t)}\| \leq \varepsilon_h$ and $\|g_v^{(t)}\| \leq \varepsilon_g$, then

$$\|[X_i, X_j](\theta)\| \leq 2H \varepsilon_g + 2G_{\max} \varepsilon_h,$$

with H bounding within-block Hessians and $G_{\max}$ bounding gradient norms. This makes ε_h (a mixed-block Hessian norm) and ε_g (out-of-support gradient size) directly *estimable* via commutator surrogates under mild boundedness assumptions.

How to use this for LLC perturbation (text). Given a modularization and a context t , use either: (i) direct mixed-Hessian measurements (exact Hessian blocks, Gauss-Newton/Fisher surrogates), or (ii) commutator surrogates calibrated by bounds on within-block Hessians and gradient norms, to upper bound ρ (or the ρ_{mn}). Separately, estimate within-module curvature lower bounds μ_m (or a PL/strong-convexity proxy restricted to each module). If the computable dominance condition (23) holds with some $\eta < 1$, then the interaction remainder is dominated and the LLC is unchanged, i.e. $\Delta\lambda = 0$.

Remark (distribution shift trigger). Under distribution shift, the same monitoring quantities (mixed blocks / commutator surrogates) can move so that (23) fails. In the systems picture, that is a precise, checkable condition for “approximate factorization has broken” and thus for when a refactorization (or re-gating / re-pruning) step is provably needed.